

An exact counterexample to odd-order Hankel H-eigenvalue inheritance

Matthew J. Colbrook

11 September 2026

Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Cambridge, United Kingdom. Email: m.colbrook@damtp.cam.ac.uk.

This argument was developed with AI assistance at the author's request. Independent verification is recorded separately; no novelty or external human peer-review claim is made.

Counterexample

Use the canonical TR-15 parameters $m = 3$, $q = 2$ and $n = 2$, and the common generating vector

$$(h_0, h_1, \dots, h_6) = (2, 0, 1, 0, 2, 0, -1).$$

Let A be the order-three, dimension-three Hankel tensor and B the order-six, dimension-two Hankel tensor with entries

$$A_{ijk} = h_{i+j+k-3}, \quad B_{i_1 \dots i_6} = h_{i_1 + \dots + i_6 - 6}.$$

Their dimensions satisfy $\dim A = q(n-1) + 1$ and $\dim B = n$, and both use all seven entries of the same generating vector.

Proposition. Every real H-eigenvalue of A is strictly positive, but -1 is a real H-eigenvalue of B . Consequently the universal inheritance conjecture in TR-15 is false.

Proof. For a real vector $x = (x_1, x_2, x_3)$, the first coordinate of the order-three contraction is

$$(Ax^2)_1 = 2x_1^2 + x_2^2 + 2x_1x_3 + 2x_3^2 = (x_1 + x_3)^2 + x_1^2 + x_2^2 + x_3^2.$$

This is strictly positive for every nonzero real x . If (λ, x) is any real H-eigenpair, its first equation is $(Ax^2)_1 = \lambda x_1^2$. Thus $x_1 \neq 0$ and $\lambda > 0$. In particular there is no negative H-eigenvalue; no numerical enumeration of eigenpairs is involved.

For $v = (0, 1)$, the only nonzero product in each component of Bv^5 has its last five indices equal to two. Hence

$$Bv^5 = (h_5, h_6) = (0, -1) = -1(v_1^5, v_2^5).$$

This gives the asserted negative H-eigenvalue with an exact nonzero real eigenvector. All parameters are admissible in the canonical conjecture, proving its negation. $\square$

The premise is not vacuous

Although unnecessary for the counterexample, A does have real H-eigenvalues. Its remaining contractions are

$$(Ax^2)_2 = 2x_1x_2 + 4x_2x_3, \quad (Ax^2)_3 = x_1^2 + 2x_2^2 + 4x_1x_3 - x_3^2.$$

Set $x = (1, 0, t)$ and $\lambda = 2 + 2t + 2t^2$. The second eigen-equation is automatic and the third is equivalent to

$$2t^4 + 2t^3 + 3t^2 - 4t - 1 = 0.$$

This polynomial is -1 at zero and 2 at one, so it has a real root $0 < t < 1$. At that root the displayed x, λ are an H-eigenpair and $\lambda > 0$.

Scope and provenance

This disproves inheritance from odd lower order in the precise H-eigenvalue sense. It does not contradict the even-lower-order theorem or inheritance from a positive-semidefinite associated Hankel matrix. Indeed the associated 4×4 Hankel matrix has bottom-right entry $h_6 = -1$, so that stronger positive-semidefinite hypothesis fails. The tensor B also takes the value -1 on the polynomial evaluation at $(0, 1)$, consistently with its negative H-eigenvalue.

The prior incomplete tensor archives contained only a qualitative counterexample summary, with no generating vector or proof. The explicit coefficients and proof above were derived anew in this research pass; no absent manuscript or code is represented as recovered.

References

1. W. Ding, L. Qi and Y. Wei, *Inheritance properties and sum-of-squares decomposition of Hankel tensors: theory and algorithms*, BIT Numerical Mathematics 57 (2017), 169-190, final Section 4, concluding odd-order inheritance conjecture; Section 2 fixes the common-vector conventions. Author's journal PDF, DOI.
2. Canonical TR-15 statement, checked 11 September 2026.